

EE-2003

Computer Organization and Assembly Language

Week 16 – Lecture 31-32

Department of Computer Science/Software Engineering/ Artificial Intelligence
FAST-NUCES Peshawar

Outline

- 1 Quick Review
- 2 Introduction to Delay Loops
- 3 Factors Affecting Delay
- 4 DELAY2 Routine (8088 Assembly)
- 5 Why Calculate Delay?
- 6 Now We Say Goodbye to **EE-2003 Computer Organization and Assembly Language!**

Quick Review: What was in the Last lecture?

- Interrupts & its types

Introduction to Delay Loops

- Delay loops create fixed time delays in 8086 assembly.
- Used for:
 - Hardware interfacing
 - Display timing
 - Synchronizing I/O operations
- Delay depends on:
 - CPU clock frequency
 - Instruction cycle counts
 - Loop nesting depth

Factors Affecting Delay

- ① CPU clock frequency (e.g., 5–10 MHz)
- ② Instruction cycle timings
- ③ Jump taken vs. not taken cycles
- ④ Total iterations of nested loops

DELAY2 Routine (8088 Assembly)

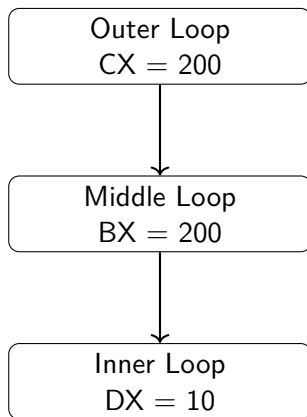
```
DELAY2:
    MOV CX, 200          ; Outer loop
AGAIN3:
    MOV BX, 200          ; Middle loop
AGAIN2:
    MOV DX, 10           ; Inner loop
AGAIN1:
    DEC DX
    JNZ AGAIN1
    DEC BX
    JNZ AGAIN2
    DEC CX
    JNZ AGAIN3
    RET
```

Total Delay Calculation for DELAY2

Total Delay Calculation (8086 @ 4.77 MHz, 1 cycle = 210 ns):

Inner loop (DX = 10)	198 cycles
Middle loop (BX = 200)	39,600 cycles
Outer loop (CX = 200)	7,920,000 cycles
RET instruction	20 cycles
Total cycles	7,920,020 cycles
Delay per cycle	210 ns
Total Delay	1.66 seconds

Nested Loop Diagram



Why Calculate Delay?

- Critical for real-time hardware interfacing.
- Ensures proper synchronization with peripherals.
- Needed in:
 - Serial communication timing
 - Display scrolling
 - Mechanical control systems
- Students understand the importance of instruction cycles.

Does Instruction Size Matter in Delay Calculations?

Short Answer: Yes — but indirectly.

- The delay of a loop depends on the **number of clock cycles** each instruction takes.
- Instruction size (1, 2, or 3 bytes) affects how many cycles the CPU needs to **fetch and execute** the instruction.
- Therefore, although delay is not computed from bytes, instruction size influences the total delay **because it changes the cycle count**.

Example (8086):

- DEC DX — 2 bytes, 2 cycles
- JNZ AGAIN1 — 2 bytes, 16 cycles (jump taken), 4 cycles (not taken)

Key Points for Students:

- Register instructions are faster than memory instructions.
- Conditional jumps have different cycles when taken vs. not taken.
- Bigger or more complex instructions usually take more cycles.
- Always use the official 8086 Instruction Cycle Table.

Thank You!

Any Questions Before We Say Goodbye!

- We explored the journey from **computer organization & Architecture** to **8088 assembly programming**.
- You learned how the CPU, memory, stack, addressing modes, subroutines, interrupts, and I/O work at the machine level.
- We built real assembly programs, debugged them, optimized them, and even created **precise delay loops**.
- And yes... you even survived segmentation, flags, calls, BIOS/DOS interrupts, and calculating delays!)

“If you can write assembly, all other programming looks easy.”

**Thank you for your patience, your questions, and for
bothering me all semester!**

Good luck in the finals!